16 Four biased coins, A, B, C and D are thrown.

The probability that each coin will land on Heads is shown in the table.

Coin	Probability
A	0.33
B	0.033
C	$\frac{1}{3}$
D	30%

(a) (i) Which coin is least likely to land on Heads?

$0.033 < 0.30 < 0.33 < 0.333...$

B

1 mark
B has the smallest probability.

(1)

(ii) Which coin is most likely to land on Heads?

$C = \frac{1}{3} = 0.333...$, the largest.

C

1 mark
C has the greatest probability.

(1)

Julie says,

"The probability that coin C will land on Heads is the same as the probability that coin C will land on Tails."

(b) Is she correct?
Give a reason for your answer.

No. $P(\text{Heads}) = \frac{1}{3}$ but $P(\text{Tails}) = 1 - \frac{1}{3} = \frac{2}{3}$

These probabilities are different.

1 mark
No, with a correct reason.

(1)

Coin B is going to be thrown 4000 times.

(c) Work out an estimate for the number of times coin B will land on Heads.

Expected Heads = $4000 \times 0.033 = 132$

1 mark
Multiply trials by probability.

1 mark
132 times.

(2)

(Total for Question 16 is 5 marks)